Communication

Theoretical analysis of non-invasive oscillometric maximum amplitude algorithm for estimating mean blood pressure

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Abstract—*A theoretical analysis is performed to evaluate the effect of arterial mechanical and blood pressure pulse properties on the accuracy of non-invasive oscillometric maximum amplitude algorithm (MAA) estimates of the mean blood pressure obtained using air-filled occlusive cuffs. Invasively recorded blood pressure pulses, selected for their varied shapes, are scaled to simulate a wide range of blood pulse pressures (diastolic blood pressure minus systolic blood pressure). Each scaled blood pressure pulse is transformed through an exponential model of an artery to create a series of blood volume pulses from which a simulated oscillometric waveform is created and the corresponding MAA estimate of the mean blood pressure and error (mean blood pressure minus MAA estimate) are determined. The MAA estimates are found to depend on the arterial blood pressure. The errors are found to depend on the arterial mechanical properties, blood pressure pulse shape and blood pulse pressure. These results suggest that there is no direct relationship between the mean blood pressure and MAA estimate, and that multiple variables may affect the accuracy of MAA estimates of the mean blood pressure obtained using air-filled occlusive cuffs.*

Keywords—*Arterial compliance, Arterial pressure–volume relationship, Maximum amplitude algorithm, Mean blood pressure, Non-invasive blood pressure, Occlusive cuff, Oscillometric method*

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1 Introduction

THE oscillometric method was first described by Marey in the late 1800s, several decades before Korotkoff's description of the auscultatory method (POSEY, *et al.*, 1969; GEDDES, 1970). However, the simpler auscultatory method eventually became the clinical standard for manual non-invasive blood pressure estimation. Over the last few decades, in parallel to the development of automated non-invasive blood pressure monitors, there has been renewed interest in the oscillometric method. Oscillometric algorithms are easily automated and are believed to provide accurate estimates of not only the diastolic and systolic blood pressure, but also, uniquely, the mean blood pressure.

The maximum amplitude algorithm (MAA) is a popular oscillometric algorithm for estimating the mean blood pressure (POSEY *et al.*, 1969; GEDDES, 1970; RAMSEY, 1979). Clinical analyses have shown that the mean difference between invasive measurements and corresponding MAA estimates of the mean blood pressure is small, approximately 5 mmHg or less (RAMSEY, 1979; YELDERMAN and REAM, 1979; EPSTEIN *et al.*, 1991). However, clinical analyses have also shown that the difference between individual invasive measurements and corresponding MAA estimates of the mean blood pressure can be large, up to approximately 40% of the invasive measurement (RUTTEN *et al.*, 1986; GRAVLEE and BROCKSCHMIDT, 1990, GORBACK *et al.*, 1991).

Additionally, *in vitro* and theoretical analyses suggest that MAA estimates depend on variables other than the mean blood pressure, including the arterial mechanical properties (URSINO and CRISTALLI, 1996), blood pressure pulse shape (FORSTER and TURNEY, 1986), blood pulse pressure (MAUCK *et al.*, 1980; YAMAKOSHI *et al.*, 1982 a; URSINO and CRISTALLI, 1996), arm tissue and cuff mechanical properties (NG and SMALL, 1993), cuff volume (MAUCK *et al.*, 1980; NG and SMALL, 1993), and pulse rate (URSINO and CRISTALLI, 1996).

This theoretical analysis is performed to evaluate further the effect of arterial mechanical and blood pressure pulse properties on the accuracy of MAA estimates of the mean blood pressure obtained using air-filled occlusive cuffs.

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2 Principles

2.1 *Oscillometric waveforms*

Figs. 1*a* and *b* show simultaneous cuff and invasive blood pressure recordings from the contralateral forelimbs of a dog, obtained during cuff inflation. Superimposed on the cuff pressure ramp are small cuff pressure oscillations that arise from blood pressure pulses in the arteries underlying the cuff. We create oscillometric waveforms by isolating and then plotting the cuff pressure oscillations or their amplitudes as a function of the cuff pressure, as shown in Figs. 1*c* and *d*.

Oscillometric waveforms can also be created by plotting blood volume pulses or their amplitudes as a function of the cuff pressure (YAMAKOSHI *et al.*, 1982 a, b; SUN, 1989). Oscillometric waveforms created from cuff pressure oscillations and blood volume pulses are similar, but not identical. To distinguish them, we call them pressure and volume oscillometric waveforms, respectively.

Volume oscillometric waveforms may provide a more direct means of estimating blood pressure. However, recording blood volume pulses from the arm, the conventional site for non-invasive blood pressure monitoring, is relatively difficult.

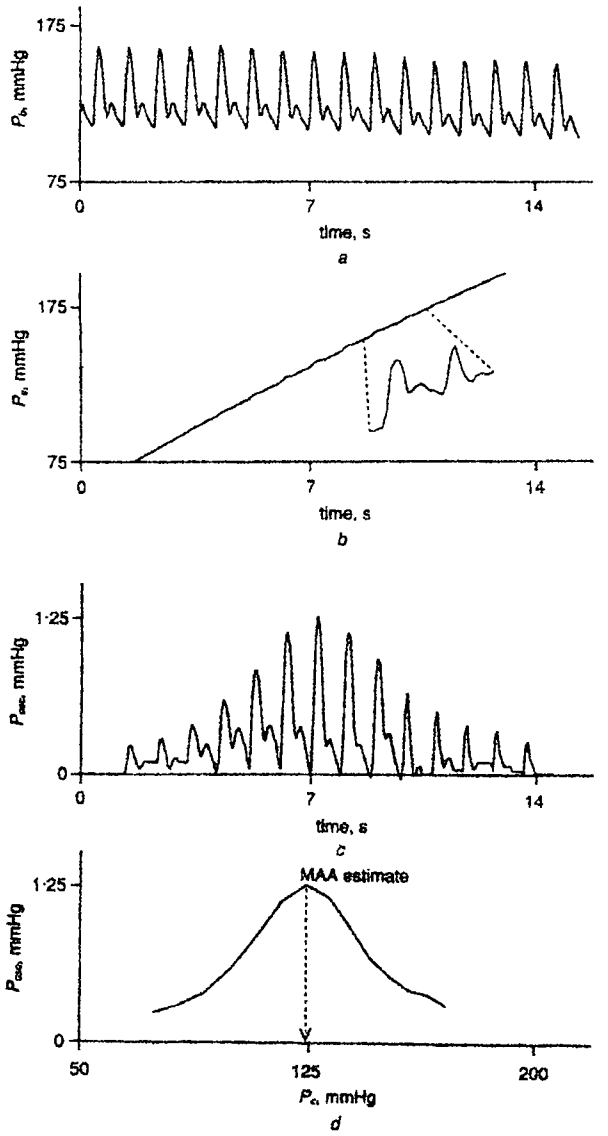

Fig. 1 *Simultaneous (a) invasive blood pressure and (b) cuff pressure recordings from the contralateral forelimbs of a dog; corresponding (c) oscillometric and (d) amplitude oscillometric waveforms; the MAA estimate is shown in (d)*

Consequently, most automated oscillometric blood pressure monitors use pressure oscillometric waveforms.

2.2 *Maximum amplitude algorithm*

Traditionally, oscillometric algorithms have been developed by empirically relating features or changes in features of the oscillometric waveform to the diastolic, mean or systolic blood pressures. Two commonly used features are the blood volume pulse or cuff pressure oscillation amplitude (POSEY *et al.*, 1969; GEDDES *et al.*, 1982; VACHTSEVANOS *et al.*, 1984) and shape (LINK, 1975; SHIMAZU *et al.*, 1986).

The MAA is an amplitude-based oscillometric algorithm. The MAA estimate of the mean blood pressure is the occlusive cuff pressure at which the blood volume pulse or occlusive cuff pressure oscillation first reach their maximum amplitude (POSEY *et al.*, 1969). One explanation for the proposed relationship between the mean blood pressure and maximum amplitude blood volume pulse or occlusive cuff pressure oscillation has been that when the occlusive cuff pressure equals the mean blood pressure the arterial wall is fully unloaded and maximally compliant, yielding the maximum change in blood volume for a given change in cuff pressure (POSEY *et al.*, 1969; DRZEWIECKI, *et al.*, 1994). However, the following theoretical anlaysis suggests that this explanation is inadequate, particularly when using air-filled occlusive cuffs.

2.3 *Theoretical analysis of oscillometric algorithms*

There have been several, fundamentally similar, approaches to the theoretical analysis of oscillometric algorithms (POSEY *et al.*, 1969; MAUCK *et al.*, 1980; GEDDES *et al.*, 1982; YAMAKOSHI *et al.*, 1982a; FORSTER and TURNEY, 1986; SUN, 1989; DRZEWIECKI *et al.*, 1994; URSINO and CRISTALLI, 1996). One of the simpler approaches is based on the concept of pulse transformation (LINK, 1975; YAMAKOSHI *et al.*, 1982 a; FORSTER and TURNEY, 1986; LINK, 1986, 1987; SUN, 1989).

Simulated blood volume pulses or cuff pressure oscillations are created by transforming blood pressure pulses (or more accurately transmural pressure pulses), through an artery or artery–tissue–cuff model, as shown in Fig. 2*a*. By repeating the transformation at different occlusive cuff pressures, which simulates occlusive cuff inflation, we create a series of simulated blood volume pulses or cuff pressure oscillations, as shown in Fig. 3*a*. Simulated oscillometric and amplitude oscillometric waveforms are then created by plotting the blood volume pulses or cuff pressure oscillations and their amplitudes, respectively, as a function of cuff pressure, as shown in Fig. 3*b*.

Although arteries contain both elastic and viscoelastic elements, their overall arterial mechanical properties appear to be dominated by the static arterial pressure–volume relationship, which has been modelled using both arctangent (GIZUDLICH and WESSELING, 1988) and exponential functions (HARDY and COLLINS, 1982; FORSTER and TURNEY, 1986; SUN, 1989; DRZEWIECKI *et al.*, 1994).

We choose an exponential model of the static arterial pressure volume relationship for this analysis:

$$V = \begin{cases} V_0 e^{aP_t}, & \text{for } P_t \leqslant 0 \\ V_{max} + (V_0 - V_{max})e^{bP_t}, & \text{for } P_t \geqslant 0 \end{cases} \quad (1)$$

where P_t is the transmural pressure, V is the blood volume at the specified P_t, V_0 is the blood volume when the artery is unloaded, V_{max} is the blood volume when the artery is fully expanded, and a and b are the exponential coefficients (SUN,

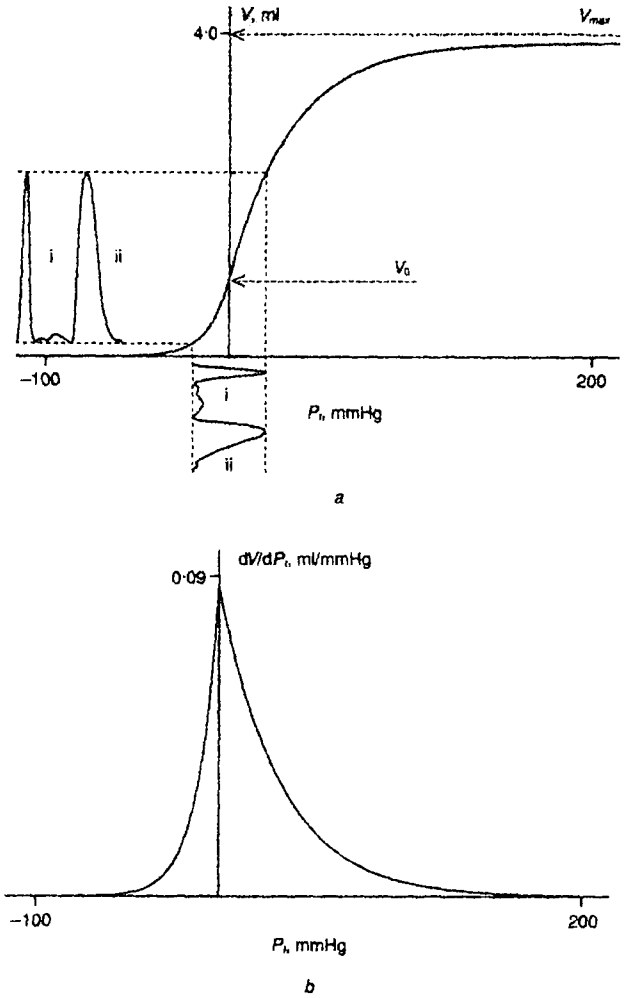

Fig. 2 *Exponential models of the static arterial (a) pressure–volume relationship and (b) compliance curve; the transformations of blood pressure pulses with blood pressure pulse shape indices of (i) 0.22 and (ii) 0.44 are shown in (a)*

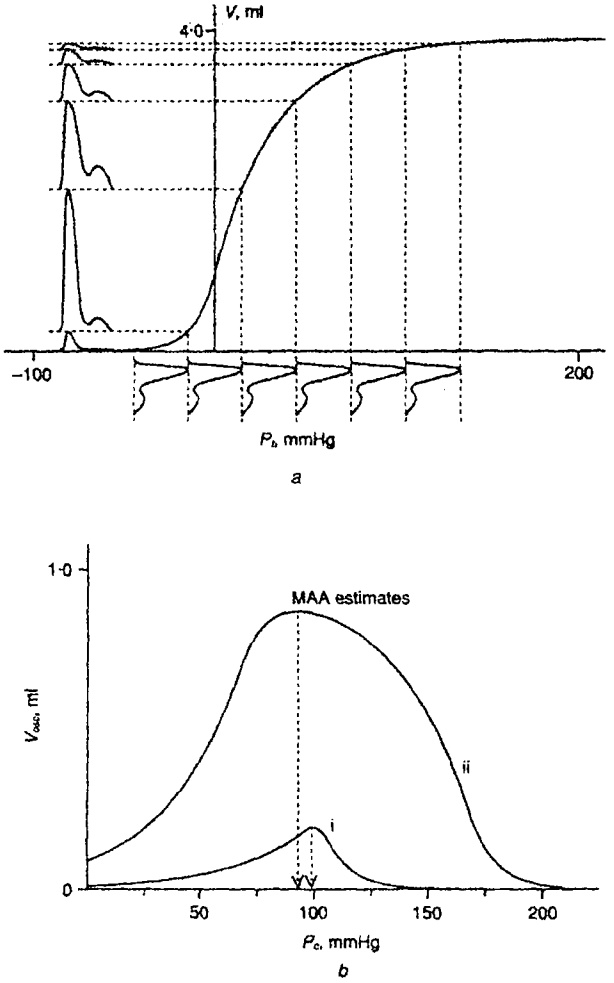

Fig. 3 *(a) Static arterial pressure–volume curve model with transformation of a blood pressure pulse at different transmural pressures, simulating different cuff pressures; (b) simulated oscillometric waveforms created from a blood pressure pulse ($k = 0.32$) scaled to blood pulse pressures of (i) 10 and (ii) 100 mmHg; the mean blood pressures of the scaled blood pressure pulses are identical; the MAA estimates, which are not identical, are shown in (b)*

1989). The first derivative of the static arterial pressure–volume curve, is the static arterial compliance

$$\frac{dV}{dP_t} = \begin{cases} aV_0 e^{aP_t}, & \text{for } P_t \leqslant 0 \\ b(V_0 - V_{max})e^{bP_t}, & \text{for } P_t \geqslant 0 \end{cases} \quad (2)$$

In this model the maximum compliance occurs at zero transmural pressure; amplitude coefficients aV_0 and $b(V_0 - V_{max})$ are equal, $a \geqslant 0$ and $b \leqslant 0$.

Theoretical analyses have shown that oscillometric waveforms created using pulse transformation and an exponential model of the static arterial pressure–volume relationship are similar in form to actual oscillometric waveforms (FORSTER and TURNEY, 1986; SUN, 1989).

3 Methods

After having obtained Institutional Animal Care and Use Committee (IACUC) approval, we recorded invasive blood pressure pulses from five anaesthetised dogs. We induced anaesthesia with sodium thiopental administered by intravenous infusion. Each dog was placed supine, intubated and mechanically ventilated to maintain the arterial partial pressure of CO_2 at approximately 35 mmHg. We maintained anaesthesia with a mixture of halothane, oxygen and nitrous oxide administered by inhalation.

We introduced a 22-gauge catheter into the radial artery of each dog through a skin incision. The catheter was connected to an invasive blood pressure monitor† through a disposable pressure transducer*. Prior to obtaining invasive blood pressure recordings from each dog, we set the pressure transducer to zero and measured the frequency response of the recording system using the flush technique (GARDNER, 1986). The natural frequency and damping coefficient of the invasive recording system ranged from 15 to 26 Hz and from 0.39 to 0.54, respectively.

We administered noradrenaline and nitroprusside by intravenous infusion, as needed, to vary the mean blood pressure of each dog between approximately 45 and 125 mmHg. Blood pressure pulses were recorded from the analogue output of the invasive blood pressure monitor, which was sampled digitally at 120 samples/s. We determined the blood pressure pulse shape index k of each blood pressure pulse

$$k = \frac{P_{bm} - P_{bd}}{P_{bs} - P_{bd}} \quad (3)$$

† Tektronix, Beaverton, Oregon, USA

* Utah Medical, Salt Lake City, USA

where P_{bd}, P_{bm} and P_{bs} are the diastolic, mean and systolic blood pressures. The blood pressure pulse shape index of the recorded pulses ranged from approximately 0.2 to 0.5. Eqn. 3 often appears in the more familiar form

$$P_{bm} = P_{bd} + k(P_{bs} - P_{bd}) \tag{4}$$

Twelve blood pressure pulses, two of which are shown in Fig. 4, were selected for their varied shapes. The blood pressure pulse shape index of the selected blood pressure pulses ranged from approximately 0.22 to 0.44 in 0.02 increments. To simulate a wide range of blood pulse pressures, each selected blood pressure pulse was scaled from 10 to 100 mmHg in 10 mmHg increments and then, for simplicity, centred about 100 mmHG by adjusting the diastolic blood pressure such that $P_{bd} = 100 - 0.5(P_{bs} - P_{bd})$. The mean blood pressure of each scaled blood pressure pulse was determined using eqn. 4.

The blood pressure pulse was assumed to approximate the transmural pressure pulse (blood pressure pulse minus cuff pressure oscillation). Thus, each scaled blood pressure pulse is transformed directly through the static arterial pressure–volume curve, at simulated cuff pressures ranging from 0 to 250 mmHg in 1 mmHG increments, to yield a series of simulated blood volume pulses, from which we create a simulated oscillometric waveform. We then determine the corresponding MAA estimate of the mean blood pressure and error (mean blood pressure minus MAA estimate).

The static arterial pressure-volume curve used in this analysis, which is shown in Fig. 2*a* was created by setting $V_{max} = 4$ ml, $V_0 = 1$ ml, $a = 0.9$ mmHg^{-1}, and $b = -0.03$ mmHg^{-1} in eqn. 1. We choose these values based on findings in the finger, where the mean ratio of V_{max} to V_0 is approximately -4, and the mean ratio of a to b is approximately 3 (SUN, 1989).

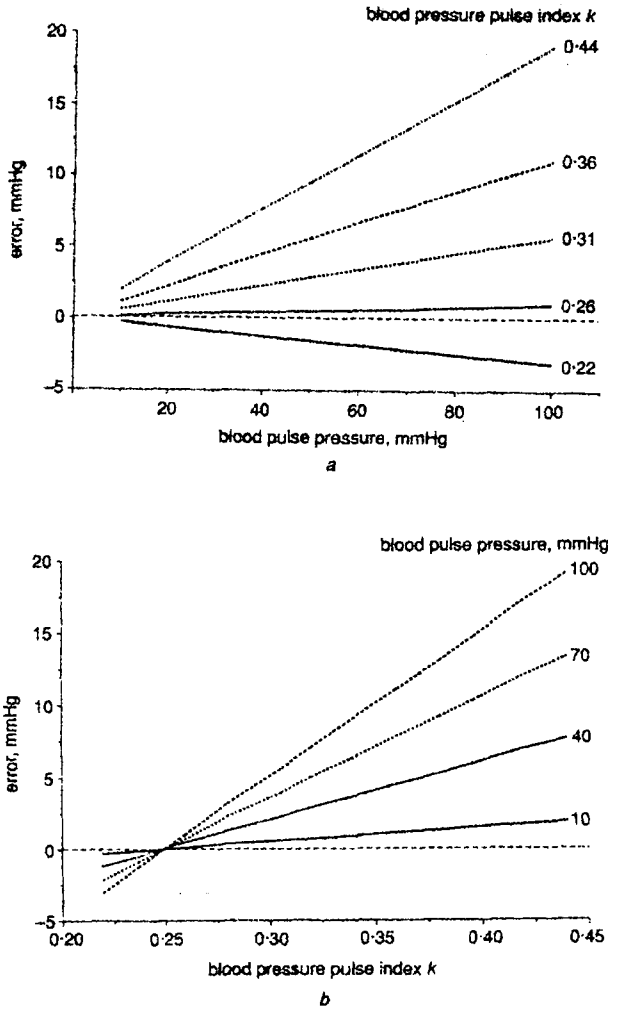

Fig. 5 *The error (mean blood pressure minus MAA estimate) as a function of (a) blood pulse pressure and (b) blood pressure pulse shape index, k*

4 Results

4.1 *Simulated oscillometric waveform results*

As shown in Figs. 5*a* and *b* we find the error (mean blood pressure minus MAA estimate) to depend on the blood pulse pressure and the pressure pulse shape index k. The maximum error is 18 mmHg, and this occurs when the blood pressure pulse shape index and blood pulse pressure are the maximum values tested, 0.44 and 100 mmHg, respectively.

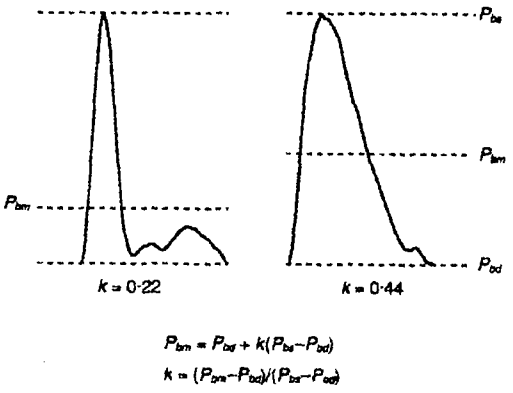

Fig. 4 *The blood pressure pulse shape index k*

4.2 *Mathematical derivation of error equation*

We derive the following error equation mathematically, as detailed in the Appendix:

$$\text{error} = \left(k + \frac{b}{a-b}\right)\Delta P_b \tag{5}$$

As with the simulated oscillometric waveforms, we find the error (mean blood pressure minus MAA estimate) to depend on the blood pressure pulse shape index k and the blood pulse pressure ΔP_b, where $\Delta P_b = P_{bs} - P_{bd}$. Additionally, we find the error to depend on the exponential coefficients, a and b, of the exponential model of the artery.

Substituting 0.09 mmHg^{-1} and -0.03 mmHg^{-1} for a and b, respectively

$$\text{error} = (k - 0.25)\Delta P_b \tag{6}$$

4.3 *Graphical anlaysis of the MAA*

The following graphical illustrations are based on the exponential model of the arterial mechanical properties described by eqns. 1 and 2. The blood pressure pulse is assumed to reasonably approximate the transmural pressure pulse.

The first illustration demonstrates the dependence of MAA estimates on the blood pulse pressure. As Fig. 2*a* shows, the transformation of blood pressure pulses with identical diastolic and systolic blood pressures yields blood volume pulses with

identical amplitudes. Thus, the resulting amplitude oscillometric waveforms and MAA estimates are identical, even though the mean blood pressures of the blood pressure pulses are different. Conversely, the transformation of blood pressure pulses with different blood pulse pressures but identical mean blood pressures yields blood volume pulses with different amplitudes. Thus, the corresponding amplitude osillometric waveforms and MAA estimates are also different, despite the mean blood pressures of the blood pressure pulses being identical.

The second illustration demonstrates the dependence of MAA estimates on the arterial mechanical properties and blood pressure pulse shape. By deduction from Fig. 6, if an arterial pressure–volume curve has a slope (arterial compliance) that is symmetric about the zero transmural pressure, the maximum amplitude blood volume pulse will occur when the transmural pressure pulse is centred on zero transmural pressure. If the blood pressure pulse shape index is 0.5, the maximum amplitude blood volume pulse occurs, fortuitously, when the cuff pressure is equal to the mean blood pressure. However, selectively changing the symmetry of the arterial compliance curve changes the shape of the amplitude oscillometric waveform and MAA estimate, even though the mean blood pressure is unchanged. Conversely, selectively changing the blood pressure pulse shape changes the mean blood pressure, even though the amplitude oscillometric waveform and MAA estimate are unchanged.

The final illustration demonstrates the conditions that yield the maximum amplitude blood volume pulse As shown in Figs. 3*a* and 6, the end-diastolic and peak-systolic transmural pressures define the segment of the arterial pressure volume curve through which the transmural pressure pulse is transformed. By deduction from these figures, the amplitude of a blood volume pulse is proportional, not to a single arterial compliance value, but to the mean arterial compliance over the segment of the arterial pressure–volume curve through which the transmural pressure pulse is transformed. Thus, the cuff pressure at which the maximum blood volume pulse occurs (MAA estimate) depends on the arterial compliance, diastolic blood pressure and blood pulse pressure (systolic minus diastolic blood pressures), but not the mean blood pressure.

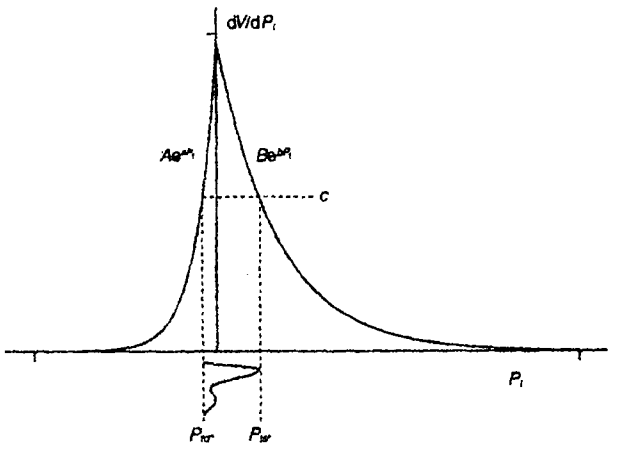

Fig. 6 *The maximum amplitude blood volume pulse occurs when the mean compliance over the segment of the static arterial pressure–volume curve model thorugh which the transmural pressure pulse is transformed is maximal; that segment, as shown, is identified by adjusting the cuff pressure such that the end-diastolic and peak-systolic transmural pressures P_{td*} and P_{ts*}, respectively, intersect the static compliance curve model at the same compliance* c, *which occurs when* $P_c = P_{bd} - P_{td*}$, *where* P_c *and* P_{bd} *are the cuff and diastolic blood pressure, respectively*

As shown in Figs. 3*a* and 6, pulse transformation is analogous to moving average filtering, where the width of the moving average filter is the transmural pulse pressure, which in this analysis is approximated by the blood pulse pressure. As the transmural pressure pulse approaches zero, the shape of the amplitude oscillometric waveform will approach that of the compliance curve. This observation has been adapted to develop sophisticated oscillometric algorithms (LINK, 1975; SHIMAZU *et al.*, 1989).

5 Discussion

5.1 *Assumptions and simplifications*

We make several assumptions and simplifications in this analysis. First, we assume the cuff pressure to be transmitted to the arterial wall without attenuation (transmural pressure equals blood pressure minus cuff pressure). Although not well demonstrated, this assumption is integral to most methods of non-invasive blood pressure estimation and it is generally considered reasonable.

Second, we assume the blood pulse pressure approximates the transmural pulse pressure (blood pulse pressure minus cuff pressure oscillation amplitude). The maximum amplitude occlusive cuff pressure oscillation in air-filled occlusive arm cuffs is usually less than 5 mmHg, in which case this assumption is reasonable. The maximum amplitude occlusive cuff pressure oscillation in fluid-filled chambers (POSEY *et al.*, 1969) or finger cuffs (YAMAKOSHI *et al.*, 1980) can approach the amplitude of the blood pressure pulse, in which case this the assumption is not reasonable and the results of this analysis are not directly applicable (the error equation would need to be modified; see eqns. A11 and A16 in the Appendix).

Third, the arm tissue and cuff mechanical properties are not included in the present model. The magnitude of the effect of the arm tissue and cuff mechanical properties on MAA estimates is unclear (MAUCK *et al.*, 1980; FORSTER and TURNEY, 1986; NG and SMALL, 1993; DRZEWIECKI, *et al.*, 1994; URSINO and CRISTALLI, 1996). Although their inclusion may alter the absolute values of the errors, they probably would not substantially affect the range or trend of the errors.

Fourth, we do not directly evaluate the effect of varying the arterial compliance. However, as shown in eqn. 5, we find the error to be proportional to the difference between the pressure pulse shape index and a constant determined by the ratio of the exponential coefficients, *a* and *b*. In the finger, the ratio of *a* to *b* ranges from approximately -1 to -6 with an average of -3 (SUN, 1989), which was the value used in this anlaysis. By deduction from eqn. 6, the use of either -1 or -6 would have increased the range of the errors in this analysis.

Fifth, we assume the static arterial pressure–volume relationship to provide a reasonable estimate of the overall arterial mechanical properties. As oscillometric waveforms created using exponential models of the static arterial pressure–volume relationship are similar to actual oscillometric waveforms in form (FORSTER and TURNEY, 1986; SUN, 1989), this assumption seems reasonable. However, a recent theoretical anlaysis suggests that MAA estimates depend on pulse rate (URSINO and CRISTALLI, 1996). Thus, under certain conditions, such as high pulse rates or large blood pulse pressures, the dynamic arterial mechanical properties may need to be incorporated in the model.

Finally, we make assumptions about the clinical ranges of the arterial compliance, blood pressure pulse shape index and blood pulse pressure and their interdependence. The ranges of the blood pressure pulse shape index and blood pulse pressure in this analysis seems reasonable. The blood pressure pulse

shape index is known to range from approximately 0.1 to 0.6 in dogs whose blood pressures are altered pharamacologically (VOELZ, 1981) and from 0.2 to 0.6 in humans in the operating room and intensive care unit (LEMLER *et al.*, 1996) and clinically the blood pulse pressure is known to range from less than 10 mmHg to greater than 100 mmHg in humans. However, the clinical range of the arterial compliance and interdependence of the arterial compliance, blood pressure pulse shape index, and blood pulse pressure are not well known. We do not vary the arterial compliance in this analysis, and the arterial compliance, blood pressure pulse shape index and blood pulse pressure are assumed to be independent variables, although they are probably not.

5.2 *Comparison with previous analyses*

Although there are several exceptions, the results of this analysis generally agree with those of the previous analyses.

First, Forster and Turney noted the effect of the blood pressure pulse shape on the error (mean blood pressure minus MAA estimate) (FORSTER and TURNEY, 1986) using an exponential model of an artery, similar to the one used in this analysis. They found that MAA estimates tend increasingly to underestimate the mean blood pressure as the blood pressure pulse shape index (quantified in terms of the fraction of the cycle that a sine wave used to simulate a blood pressure pulse was non-zero) is increased.

Second, Mauck *et al.* noted the effect of the blood pulse pressure on the error (MAUCK *et al.*, 1980) using a one-dimensional model of the artery-tissue-cuff. Yamakoshi *et al.* used excised dog carotid arteries placed in a fluid-filled chamber and perfused with sinusoidal pressure pulses (YAMAKOSHI *et al.*, 1982a). Ursino and Cristalli used a lumped parameter model of the artery-tissue-cuff (URSINO and CRISTALLI, 1996). They all found that MAA estimates tend increasingly to underestimate the mean blood pressure as the blood pulse pressure is increased. In contrast, however, Forster and Turney found that the error is not significantly affected by the blood pulse pressure (FORSTER and TURNEY, 1986).

A possible explanation for the findings of Forster and Turney is that the blood pressure pulse shape index of their simulated blood pressure pulse and constant determined by the exponential coefficients of their model may have been almost equal. Consequently, as shown in eqn. 5 and as was found by Forster and Turney, the error would appear to be independent of changes in the blood pulse pressure.

Finally, Ursino and Cristalli noted the effect of the arterial mechanical properties on the error (URSINO and CRISTALLI, 1996). They found that the MAA increasingly underestimates the mean blood pressure as the artery stiffness is increased. In contrast, however, Mauck *et al.* and Forster and Turney found that the error was not significantly affected by the arterial mechanical properties.

A possible explanation for the finding of Mauch *et al.* is that they assumed that the arterial volume was zero for transmural pressure less than zero. In this analysis that would be equivalent to setting the exponential coefficients a in eqn. 5 to zero. Consequently, as was found by Mauck *et al.*, the error would appear to be independent of the arterial mechanical properties.

A possible explanation for the finding of Forster and Turney is that they apparently varied arterial compliance by changing the slope of their arterial pressure–volume curve at zero transmural pressure, without changing the ratio of the exponential coefficients of their exponential artery model. Consequently, as shown in eqn. 5 and as was found by Forster and Turney, the error would appear to be independent of the arterial mechanical properties.

5.3 *Summary*

These results suggest that there is not a direct relationship between the mean blood pressure and the MAA estimate, and that multiple variables, including the arterial mechanical properties, blood pressure pulse shape and blood pulse pressure, may affect the accuracy of MAA estimates of the mean blood pressure obtained using air-filled occlusive cuffs.

Additional research into the clinical range and interdependence of these variables should be useful in further evaluating their effect on MAA estimates.

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References

DRZEWIECKI, G., HOOD, R., and APPLE, H. (1994): 'Theory of the oscillometric maximum and the systolic and diastolic detection ratios,' *Ann. Biomed. Eng.*, **22**, pp. 88–96

EPSTEIN, R. H., HUFFNAGLE, S., and BARTKOWSKI, R. R. (1991): 'Comparative accuracies of a finger blood pressure monitor and an oscillometric blood pressure monitor,' *J. Clin. Monit.*, **7**, pp. 161–167

FORSTER, F. K., and TURNEY, D. (1986): 'Oscillometric determination of diastolic, mean and systolic blood pressure—a numerical model,' *J. Biomech. Eng.*, **108**, pp. 359–364

GARDNER, R. M. (1986): 'Hemodynamic monitoring: from catheter to display,' *Acute Care*, **12**, pp. 3–33.

GEDDES, L. A. (1970): 'The direct and indirect measurement of blood pressure' (Year Book Medical Publishers, Chicago)

GEDDES, L. A., VOELZ, M., COMBS, C., REINER, D., and BABBS, C. F. (1982): 'Characterization of the oscillometric method for measuring indirect blood pressure,' *Ann. Biomed. Eng.*, **10**, pp. 271–280

GIZDULICH, P., and WESSELING, K. H. (1988): 'Forearm arterial pressure–volume relationships in man,' *Clin. Phys. Physiol. Meas.*, **9**, pp. 123–132

GORBACK, M. S., QUILL, T. J., and LAVINE, M. L. (1991): 'The relative accuracies of two automated noninvasive arterial pressure measurement devices,' *J. Clin. Monit.*, **7**, pp. 13–22

GRAVLEE, G. P., and BROCKSCHMIDT, J. K. (1990): 'Accuracy of four indirect methods of blood pressure measurement, with hemodynamic correlations,' *J. Clin. Monit.*, **6**, pp. 284–298

HARDY, H. H., and COLLINS, R. E. (1982): 'On the pressure–volume relationship in circulatory elements,' *Med. Biol. Eng. Comput.*, **20**, pp. 565–570

LEMLER, J., KÜCK, K., HARYADI, D. G., BAKER, P. B., and WESTERNSKOW, D. R. (1996): 'Variability of the blood pressure k "constant" in humans. *Proceedings of the Annual Meeting of the Society for Technology in Anesthesia*, San Diego, California, USA

LINK, W. T. (1975): 'Apparatus and process for producing sphygmometric information.' US Patent 3,903,872

LINK, W. T. (1986): 'Method and apparatus for obtaining an individual's systolic blood pressure.' US Patent 4,564,020

LINK, W. T. (1987): 'Techniques for obtaining information associated with an individual's blood pressure including specifically a stat mode technique.' US Patent 4,699,152

MAUCK, G. W., SMITH, C. R., GEDDES, L. A., and BOURLAND, J. D. (1980): 'The meaning of the point of maximum oscillations in cuff pressure in the indirect measurement of blood pressure—Part II,' *J. Biomech. Eng.*, **102**, pp. 28–33

NG, K.-G., and SMALL, C. F. (1993): 'Changes in oscillometric pulse amplitude envelope with cuff size: implications for blood pressure measurement criteria and cuff size selection,' *J. Biomed. Eng.*, **15**, pp. 279–282

POSEY, J. A., GEDDES, L. A., WILLIAMS, H., and MOORE, A. G. (1969): 'The meaning of the point of maximum oscillations in cuff pressure in the indirect measurement of blood pressure: Part I,' *Cardiovasc. Res. Center Bull.*, **8**, pp. 15–25

RAMSEY, M. (1979): 'Noninvasive automatic determination of mean arterial pressure,' *Med. Biol. Eng. Comput.*, **17**, pp. 11–18

RUTTEN, A. J., ILSLEY, A. H., SKOWRONSKI, G. A., and RUNCIMAN, W. B. (1986): 'A comparative study of the measurement of mean arterial blood pressure using automatic oscillometers, arterial cannulation and auscultation,' *Anaesth. Intens. Care*, **14**, pp. 58–65

SHIMAZU, H., ITO, H., KOBAYASHI, H., and YAMAKOSHI, K. (1986): 'Idea to measure diastolic arterial pressure by volume oscillometric method in human fingers,' *Med. Biol. Eng. Comput.*, **24**, pp. 549–554

SHIMAZU, H., ITO, H., KAWARADA, A., KOBAYASHI, H., HIRAIWA, A., and YAMAKOSHI, K. (1989): 'Vibration technique for indirect measurement of diastolic arterial pressure in human fingers,' *Med. Biol. Eng. Comput.*, **27**, pp. 130–136

SUN, S. (1989): 'New approaches to the noninvasive determination of arterial blood pressure and compliance profile.' PhD dissertation, University of Utah, Salt Lake City, USA

URSINO, M. and CRISTALLI, C. (1996): 'A mathematical study of some biomechanical factors affecting the oscillometric blood pressure measurement.' *IEEE Trans. Biomed. Eng.*, **43**, pp. 761–778

VACHTSEVANOS, G., KALAITZAKIS, C., PAPAMARKOS, N., ZIAKAS, G., ECONOMOU, K., and GEMITZIS, K. (1984): 'Correlation study of arterial blood pressure level to the amplitude of the pressure pulse waveform,' *J. Biomed. Eng.*, **6**, pp. 33–39

VOELZ, M. (1981): 'Measurement of the blood-pressure constant k, over a pressure range in the canine radial artery,' *Med. Biol. Eng. Comput.*, **19**, pp. 535–537

WESSELING, K. H., SETTELS, J. J., and DE WIT, B. (1986): 'The measurement of continuous finger arterial pressure noninvasively in stationary subjects' *in* Schmidt, T. H. *et al.* (Eds.): '*Biological and psychological factors in cardiovascular disease*' (Springer-Verlag, Berlin, Heidelberg) pp. 355–375

YAMAKOSHI, K., SHIMAZU, H., and TOGAWA, T. (1980): 'Indirect measurement of instantaneous arterial blood pressure in the human finger by the vascular unloading technique,' *IEEE Trans. Biomed. Eng.*, **27**, pp. 150–155

YAMAKOSHI, K., SHIMAZU, H., SHIBATA, M., and KAMIYA, A. (1982 a): 'New oscillometric method for indirect measurement of systolic and mean arterial pressure in the human finger. Part 1: model experiment,' *Med. Biol. Eng. Comput.*, **20**, pp. 307–313

YAMAKOSHI, K., SHIMAZU, H., SHIBATA, M., and KAMIYA, A. (1982 b): 'New oscillometric method for indirect measurement of systolic and mean arterial pressure in the human finger. Part 2: correlation study,' *Med. Biol. Eng. Comput.*, **20**, pp. 314–318

YELDERMAN, M., and REAM, A. K. (1979): 'Indirect measurement of mean blood pressure in the anesthetized patient, *Anesthesiology*, **50**, pp. 253–256

Appendix

The following is a derivation of the error equation, which describes the difference between the mean blood pressure and maximum amplitude algorithm (MAA) estimate.

Notation

P_b = blood pressure
P_{bd} = diastolic blood pressure
P_{bm} = mean blood pressure
P_{bs} = systolic blood pressure
P_{bm^*} = maximum amplitude algorithm estimate of mean blood pressure
P_c = cuff pressure (assumed to be approximately constant during a blood pressure pulse)
k = blood pressure pulse shape index
P_t = transmural pressure
P_{td} = P_t at end-diastole
P_{ts} = P_t at peak-systole
P_{td^*} = P_{td} corresponding to the maximum blood volume pulse
P_{ts^*} = P_{ts} corresponding to the maximum blood volume pulse
ΔP_b = blood pulse pressure
ΔP_t = transmural pulse pressure
a, b = exponential coefficients of the artery model
A, B = amplitude coefficients of the artery model
c = arterial compliance corresponding to P_{td^*} and P_{ts^*}
error = difference between the mean blood pressure and MAA estimate

Identities

$$k = \frac{P_{bm} - P_{bd}}{P_{bs} - P_{bd}}$$

$$P_t = P_b - P_c$$

$$\Delta P_b = P_{bs} - P_{bd}$$

$$\Delta P_t = P_{ts} - P_{td}$$

The derivation of the error equation begins with a simplified version of the exponential model of the static arterial compliance used in this analysis:

$$\frac{dV}{dP_t} = \begin{cases} Ae^{aP_t}, & \text{for } P_t \leqslant 0 \\ Be^{bP_t}, & \text{for } P_t \geqslant 0 \end{cases} \tag{A1}$$

and the observation that the maximum amplitude blood volume pulse occurs when the mean compliance over the segment of the static arterial pressure-volume model, through which the transmural pressure pulse is transformed, is maximal. As shown in Fig. 6, we identify that segment by adjusting the cuff pressure such that the end-diastolic and peak-systolic transmural pressures, P_{td^*} and P_{ts^*}, respectively, intersect the compliance curve at the same compliance c. Thus, the maximum amplitude blood volume pulse occurs when

$$c = Ae^{aP_{td^*}} = Be^{bP_{ts^*}} \tag{A2}$$

$$P_{td^*} = \frac{1}{a}\ln\left(\frac{c}{A}\right) \tag{A3}$$

$$P_{ts^*} = \frac{1}{b}\ln\left(\frac{c}{B}\right) \tag{A4}$$

By definition

$$\Delta P_t = P_{ts^*} - P_{td^*} \tag{A5}$$

By substituting eqns. A3 and A4 into A5

$$\Delta P_t = \left(\frac{1}{b} - \frac{1}{a}\right)\ln(c) - \frac{1}{b}\ln(B) + \frac{1}{a}\ln(A) \tag{A6}$$

Rearranging

$$\ln(c) = \frac{ab}{a-b}\left(\Delta P_t + \frac{1}{b}\ln(B) - \frac{1}{a}\ln(A)\right) \tag{A7}$$

By definition

$$\text{error} = P_{bm} - P_{bm^*} \tag{A8}$$

By deduction from Fig. 6

$$P_{bm^*} = P_c = P_{bd} - P_{td^*} \tag{A9}$$

and by definition

$$P_{bm} = P_{bd} + k(P_{bs} - P_{bd}) = P_{bd} + k\Delta P_b \tag{A10}$$

By substituting eqns. A9 and A10 into eqn. A8

$$\text{error} = k\Delta P_b + P_{td^*} \tag{A11}$$

Assuming $\Delta P_t = \Delta P_b$, which is reasonable when the cuff is relatively compliant

$$\text{error} = k\Delta P_t + P_{td^*} \tag{A12}$$

By substituting eqns. A3 and A6 into eqn. A12 and rearranging

$$\text{error} = k\left[\left(\frac{1}{b} - \frac{1}{a}\right)\ln(c) - \frac{1}{b}\ln(B) + \frac{1}{a}\ln(A)\right] + \frac{1}{a}\ln(c) - \frac{1}{a}\ln(A) \quad \text{(A13)}$$

By substituting eqn. A7 into A13 and rearranging

$$\text{error} = \left(k + \frac{b}{a-b}\right)\Delta P_t + \frac{1}{a-b}(\ln(B) - \ln(A)) \quad \text{(A14)}$$

By deduction from eqn. A1, $A = B$; thus

$$\text{error} = \left(k + \frac{b}{a-b}\right)\Delta P_t \quad \text{(A15)}$$

Again, by assuming, $\Delta P_t = \Delta P_b$, which is reasonable when the cuff is relatively compliant

$$\text{error} = \left(k + \frac{b}{a-b}\right)\Delta P_b \quad \text{(A16)}$$